MIL-T-45695(Ord)
14 April 1961
SUPERSEDING
FA(FC)-PD-111
16 June 1959

MILITARY SPECIFICATION

TELESCOPE, STRAIGHT: XM106

1. SCOPE

1.1 This specification covers one type of telescope known as Telescope, Straight XM106, hereinafter known as telescope, used in conjunction with radar target and missile trackers.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical components for Fire Control Instruments; General Specifications Governing the Manufacture Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14619	Fire Control Parts Equipment and Tools, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Ordnance Corps

F8270944	Device, Special Testing
F8614825	Telescope, Straight: XM106
LSQAP 8614825	List of Supplementary Quality Assurance Provisions

FSC 1240

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one telescope, representative of production, shall be made available to the Government to determine if the telescope meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The telescope shall be manufactured in accordance with Drawing F8614825, "Telescope, Straight, XM106" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-O-13830 and MIL-F-13926:

3.4.1 MIL-O-13830

(a) Cleanliness
(b) Illuminated reticle

3.4.2 MIL-F-13926.

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Storage temperature.- The telescope shall be exposed and thermally stabilized at ambient temperatures of minus 80 degrees and plus 160 degrees Fahrenheit. Upon return to standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) the telescope shall show no indication of physical failure and shall meet the performance requirements of this specification.

3.5.2 Operating temperature.- The telescope shall meet the maximum torque requirements of 3.14.1 and 3.14.2 when exposed and thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit, subsequent to which, the telescope shall meet the requirements of 3.5.3 at standard ambient temperature.

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3.5.3 Vibration.- With the telescope supported by the cylindrical surfaces of its mounting bearings, the telescope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration the telescope shall not be damaged, there shall be no broken, loose or missing parts, and the telescope shall meet the requirements of 3.6 to 3.14 inclusive.

3.6 Cleanliness.- Subsequent to vibration the telescope shall meet the cleanliness requirements of Specification MIL-O-13830 and there shall be no more than 2 particles of dirt visible on the reticle surfaces, and each particle shall not be in excess of an area equivalent to a size 10 dig. There shall be no dirt on other surfaces whichinterferes with observation or adversely affects image quality.

3.7 Sealing.- The telescope shall show no evidence of leakage in excess of 0.05 psig when subjected to an internal pressure of dry nitrogen at 5 psig for 5 minutes. The internal atmosphere of the accepted telescope shall contain nitrogen (dew-point of nitrogen shall be at least as low as minus 25 degrees Fahrenheit) at a pressure of between 0.1 and 0.25 psig.

3.8 Collimation.- With the telescope supported on its cylindrical mounting surfaces and the locating keyway of the rear mounting bearing on top (at twelve o'clock position) and the line of sight through the optical axis of the telescope horizontal, the intersecting point of the reticle lines when focused on a target point shall not deviate from the target point in excess of 0.025 mil radius when the telescope is rotated on its cylindrical mounting surfaces, and focused over the extremes specified in 3.10.

3.8.1 Level vial position.- The level vial bubble shall be central within the width of a vial graduation line when the locating keyway is on top.

3.8.2 Level vial roll.- Rotation of the telescope on its cylindrical mounting surfaces to positions of 20 degrees clockwise and 20 degrees counterclockwise from the position established in 3.8.1 shall not cause the level bubble to move by more than a total of one division (space) of the vial.

3.8.3 Reticle alinement.- The vertical reticle line shall not deviate from a vertical target line (plumb line) by more than an angle of one degree, when the locating keyway is on top.

3.9 Eyepiece focus.- With the crosslines of the reticle in sharp focus, as viewed through an auxiliary telescope, the eyepiece diopter scale shall read zero diopters, from which position the diopter scale shall be adjustable over a range of plus or minus three diopters.

3.10 Image focus.- Rotation of the focusing adjustment shall bring an image into sharp focus when viewed at any range between seventy meters and infinity.

3.11 Resolution.- The resolution on the optical axis shall be 5 seconds of arc or less, when focused over the extremes specified in 3.10.

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3.12 Field of view.- The telescope shall have a minimum field of view of 55 minutes.

3.13 Reticle illumination.- The reticle when illuminated by the telescope instrument light and checked throughout the total range of focus of the telescope, shall be clearly defined when viewed in a darkened area.

3.14 Operability.

3.14.1 Diopter torque.- The minimum torque required to rotate the diopter adjustment shall not be less than 5 ounce inches when checked at standard ambient temperature. The maximum torque required to rotate the diopter adjustment shall not exceed 3 pound inches when checked to the requirements of 3.5.2. The diopter adjustment shall be operable when checked at extreme temperatures of 3.5.1.

3.14.2 Focusing torque.- The minimum torque required to rotate the focusing collar shall not be less than 5 ounce inches when checked at standard ambient temperature. The maximum torque required to rotate the focusing collar shall not exceed 8 pound inches when checked to requirements of 3.5.2. The focusing adjustment shall be operable when checked at extreme temperatures of 3.5.1.

3.15 Identification marking.- Identification marking unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality.

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The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Classification of tests.- The inspection and testing of the telescope shall be classified as follows:

(a) Production model test
(b) Acceptance test

4.3 Production model.

4.3.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

4.3.2 These tests shall be conducted at the contractor's facility using test equipment designed or procured for use on the remaining items.

4.3.3 The model shall meet all the requirements of this specification using the inspection methods authorized herein.

4.3.3.1 Disposition of rejected production model.- Should the production model fail to pass the test for Government acceptance, production of these components causing rejection will cease until the defective characteristics have been corrected in the production process. When corrections in the process have been made, a new production model shall be completely inspected by the Government. The rejected production model may be corrected, where practicable, and shall become part of the first production lot.

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4.4 Inspection provisions.

4.4.1 Inspection of components.- Inspection requirements for component parts and sub-assemblies are contained herein and on SQAP's listed on LSQAP 8614825.

4.4.1.1 The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items.

4.4.1.1.1 Contractor's inspection for all components not covered by SQAP shall be in accordance with the inspection procedures of the written inspection plan required by 4.1.1. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements that have a bearing on the functional accuracy and performance of the Telescope, Straight: XM106.

4.4.2 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitable equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.4.3 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.3.1 Acceptance tests.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items. Samples selected for acceptance test shall include the sample selected for environmental testing.

4.4.3.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE I - CLASSIFICATION OF DEFECTS

For Normal Inspection Level L7, and Sampling Plan Table IV-A
of Standard MIL-STD-105
The AQL value is expressed in percent defective

MAJOR: AQL 6.5	Req't Par.	Q.A. Par.	Insp. Equipment or Method
101. Storage temperature	3.5.1	4.5.1	Standard Measuring Equipment (SME)
102. Operating temperature	3.5.2	4.5.1.1	SME
103. Vibration	3.5.3	4.5.2	SME
104. Sealing	3.7	4.5.4	SME

TABLE II - CLASSIFICATION OF DEFECTS

For Normal Inspection Level II, and Sampling Plan Table IV-A
of Standard MIL-STD-105
For Reduced Inspection Level I, and Sampling Plan Table IV-A
of Standard MIL-STD-105
The AQL values are expressed in percent defective

MAJOR: AQL 1.0	Req't Par.	Q.A. Par.	Insp. Equipment or Method
101. Collimation	3.8	4.5.5.1	Device, Special Testing (DST)
102. Level vial position	3.8.1	4.5.6	DST
103. Level vial roll	3.8.2	4.5.6.1	DST
104. Reticle alinement	3.8.3	4.5.7	DST
105. Eyepiece focus	3.9	4.5.8	DST
106. Image focus	3.10	4.5.9	DST
107. Definition	3.11	4.5.10	DST
108. Field of view	3.12	4.5.11	DST

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TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

MINOR: AQL 2.5	Req't Par.	Q.A. Par.	Insp. Equipment or Method
201. Reticle illumination	3.13	4.5.12	Visual
202. Diopter adjustment	3.14.1	4.5.13	SME
203. Focusing adjustment	3.14.2	4.5.13.1	SME
204. Cleanliness	3.6	4.5.3	Visual
205. Identification marking	3.15	4.5.14	Visual

4.4.4 Inspection equipment.

4.4.4.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection, equipment, and to use it for Government verification inspection or acceptance inspection.

4.4.4.2 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.5 Test methods and procedures.

4.5.1 Storage temperature.- The telescope shall be installed in the test chamber for tests. The equipment under test shall not exceed fifty (50) percent of the internal volume of the test chamber and no portion shall be closer than six (6) inches from any surface of the chamber. The telescope under test shall be placed in the test chamber, and the temperature shall be thermally stabilized in accordance with 3.5.1. Upon completion of the test, the telescope shall be inspected to determine compliance with the test procedures outlined herein.

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4.5.1.1 Operating temperature.- The temperature of the test chamber shall be thermally stabilized in accordance with 3.5.2. The telescope shall be exposed to these temperatures and shall be inspected to determine compliance with the requirements of 3.5.2.

4.5.2 Vibration.- The telescope shall be mounted on a vibration table by means of an adapter. The telescope shall then be vibrated at the amplitude, frequency and duration specified in 3.5.3. After the vibration test, the telescope shall be inspected as outlined below to determine compliance with the requirements of 3.5.3.

4.5.3 Cleanliness.- The cleanliness test shall be performed in accordance with the shadowing method outlined in MIL-O-13830. Dirt particles shall not exceed the limits specified in 3.6.

4.5.4 Sealing.- The telescope shall be attached to an approved testing device and shall be internally pressure tested for compliance with the requirements of 3.7.

4.5.5 The following test outlined herein is performed using a special testing device shown on Drawing F8270944. This drawing may be obtained through the contracting officer and will be used for informational purposes only to demonstrate the inspection method rather than the accuracy of the special testing device.

4.5.5.1 Collimation.- Upon completion of the above, the diopter scale of the telescope under test shall be set to indicate zero. The eyepiece shall be adjusted to produce a clear sharp image of the reticle when viewed against a white background. The telescope shall be properly mounted, on the "V" supports of the fixture, as far forward as possible to clear the locating key of the fixture. The telescope's focusing collar shall be adjusted to produce a clear picture of the collimator target, when the collimator is set at the minimum distance specified in 3.10. This is to be accomplished without moving the telescope's eyepiece focus. Observe the position of the intersecting point of the instrument's reticle with respect to the target circle, which represents a diameter of .2 mil. The telescope's intersecting point of the reticle must be in coincidence with the optical axis of the collimator within the limits of the circle. When this alinement has been met, the collimation requirement is checked by rotating the telescope on the mounting surfaces and observing the amount of movement between the intersecting point of the telescope's reticle and the target point. This movement shall not exceed .10 mil. This movement shall not exceed 50 percent of the target circle. Repeat this procedure at infinity setting of the collimator. This will require a re-setting of the telescope's focusing scale.

4.5.6 Level vial position.- With the telescope positioned on the fixture with the keyway in the twelve (12) o'clock position and the optical axis concentric with the mounting surfaces, the level bubble shall be central within the limits specified in 3.8.1.

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4.5.6.1 Level vial roll.- With the telescope's level vial positioned as mentioned above, rotate the telescope on the mounting surfaces to position the level vial twenty (20) degrees each side of this established position, as determined by the fixture's limit stops, the bubble movement level vial shall not exceed the requirements specified in 3.8.2.

4.5.7 Reticle alinement.- Is accomplished by positioning the telescope to engage the key located in the "V" block with the keyway in the telescope's collar and observing parallelism between the telescope's vertical reticle line and the target vertical line. Proper reticle alinement is assured when the reticle vertical line is parallel to the target's vertical line within the limits specified in 3.8.3. Tolerance lines are provided on the collimator reticle to aid in determining reticle alinement.

4.5.8 Eyepiece focus.- Is accomplished by utilizing an auxiliary telescope of 4X power. Prior to performing this test, the auxiliary telescope must be set as follows: Adjust the eyepiece to produce a sharp image of the reticle when viewing against a white background. Next focus the auxiliary telescope on a target, such as a smoke stack, edge of a building, etc. at a target distance of 500 yards to remove parallax. Parallax is removed when there is no displacement between the reticle and target when the observer's head is moved from side to side or up and down. Set the telescope diopter scales to zero and position the auxiliary telescope behind the eyepiece assembly, observe the reticle image of the telescope under test. Obtain a clear sharp image of the telescope's reticle by rotating the diopter scale of the auxiliary telescope. The diopter scale setting shall be within the limits specified in 3.9 and the telescope shall be capable of producing an adjustment as specified in 3.9.

4.5.9 Image focus.- The telescope is properly focused when parallax has been removed. In order to perform this test, the collimator will be set to 70 meters. Adjust the telescope's range ring to produce a clear sharp image of the collimator reticle. The actual check is made by observing parallax between the collimator reticle and the telescope's reticle. Proper image focus is assured when there is no movement between a corresponding point on the collimator reticle with respect to the corresponding point of the telescope's reticle in close proximity of the telescope's optical axis. Parallax is recognized as displacement between the points when the observer's head is moved from side to side or up and down. Repeat the procedure at the infinity setting of the collimator.

4.5.10 Resolution.- The five (5) seconds of arc resolving power specified in 3.11 shall be checked on the final test fixture. The test pattern, when viewed through the telescope shall be resolved in each of the four (4) meridians; by a movement of the telescope's diopter scale not in excess of plus or minus 1/8 diopter for the four (4) meridians. The test pattern shall be considered resolved, when the correct line count can be determined in all four (4) meridians.

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4.5.11 Field of view.- The 55 minutes field of view specified in 3.12 shall be determined by observing the properly calibrated horizontal and vertical lines of the collimator reticle. The field of view shall be acceptable when the 55 minutes tolerance lines of the collimator are clearly observed.

4.5.12 Reticle illumination.- The telescope's instrument light shall be attached to the instrument. The horizontal and vertical reticle lines shall appear clearly defined when tested in a dark room or with the objective end of the telescope covered to prevent the entrance of extraneous light. This test is to be performed over the full travel range of the reticle assembly.

4.5.13 Diopter adjustment.- The torque test shall be performed to determine that the diopter adjustment, when subjected to the temperatures specified in 3.14.1 shall show no evidence of bind. In the performance of this test, the telescope shall be secured in a holding device, such as a "V" block, etc. With a suitable adapter secured to the diopter scale and using an accurately calibrated torque wrench, the rotation of the diopter adjustment shall not exceed the limits specified in 3.14.1.

4.5.13.1 Focusing adjustment.- The telescope shall be positioned as outlined in 4.5.13 and subjected to the temperatures specified in 3.14.2. A suitable adapter shall be secured to the focusing collar. The torque required to rotate the focusing adjustment shall not exceed the limits specified in 3.14.2 as determined by an accurately calibrated torque wrench.

4.5.14 Identification marking.- Identification marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The telescope shall be packaged, packed and marked in accordance with Specification MIL-F-14619/P13045. The level of protection shall be as specified in the procurement documents.

6. NOTES

6.1 Intended use.- The Telescope, Straight: XM106 is intended for alinement, and a triangulation check of the target tracking and missile tracking radar antennas of the NIKE system, to check one antenna against the other and each against an established checking target prior to functional operation of the system.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

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6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.4 Definitions.

6.4.1 Standard measuring equipment.- Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.2 Special testing device (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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